

Slide 1 – Title

- “Good night everyone. My name is Wei JianNan, and today I will present our project: Smart Retrofit Kit for Legacy Air Conditioners. This work is registered under code MC555577.”

Slide 2 – Motivation

- “In the military domain, Russia upgraded Cold War-era FAB bombs with UMPK kits, enabling precision guidance at low cost. In the home appliance domain, legacy air conditioners lack networking and smart features. By adding a smart control gateway, we can enable remote control, intelligent adjustment, and extend lifespan. This is a cost-effective modernization.”

Slide 3 – Hardware Modules and Simulation

- “Hardware platform: MPS2-AN386, STM32F407 MCU running FreeRTOS. Modules:
 - SHT30 sensor (temperature & humidity).
 - VS1838B IR receiver (NEC protocol).
 - IR333C IR transmitter (infrared control).
 - ESP8266 Wi-Fi module. Interfaces: I2C, UART, PWM, GPIO, HTTP, JSON. Objective: Intelligent AC control gateway + QEMU + Python simulation.”

Slide 4 – System Tasks

- “Tasks:
 - Sensor Task: collect environment data.
 - IR Task: infrared AC control.
 - Control Task: merge sensor data & user input, AI suggestion.
 - Wi-Fi Task: manage ESP8266 connectivity.
 - Bridge Task: external communication. Temperature suggestion rules:
 - Spring/Autumn: outdoor $\pm 1^{\circ}\text{C}$
 - Summer: outdoor -4°C
 - Winter: outdoor $+6^{\circ}\text{C}$.”

Slide 5 – Communication Mechanisms

- “FreeRTOS queues: xSensorQueue, xCommandQueue, xStatusQueue. Mutex: protects shared resources, prevents deadlocks. Patterns: producer-consumer, synchronous messaging, asynchronous sampling.”

Slide 6 – Weather API / Wi-Fi / Web Integration

- “Python Bridge Server fetches real-time weather from Open-Meteo API (Macau). Workflow:
 1. API call $\rightarrow$ JSON response.
 2. Bridge Server packages commands.

3. Wi-Fi Task simulates ESP8266 AT commands, receives JSON.”

Slide 7 – Smart Config Implementation

- “SmartConfig avoids hardcoding Wi-Fi info. Steps:
 - User enters Wi-Fi info in mobile app.
 - App broadcasts SSID & password via UDP.
 - ESP8266 listens, decodes, connects.
 - Web interface shows ‘Connected’. Works in QEMU simulation.”

Slide 8 – Web Control Panel Demo

- “Features:
 - Live status display.
 - SmartConfig connection.
 - Indoor & outdoor temperature/humidity.
 - Auto-suggested temperature.
 - Bilingual interface (Chinese/English).
 - Continuous updates via JSON.
 - Timer & scheduling: auto-off countdown, scheduled on/off.”

Slide 9 – Closing

- “Thank you for your attention. Smart Retrofit Kit modernizes legacy ACs at low cost, extends lifespan, and

adds smart features. It provides a practical solution for smart home development.”